

Chapter 2

HTML Part 4

Images and Tables

INT150 Web Technologies (2/2025)

Sanit Sirisawatvatana

Topics

- What will you learn after finishing this class:
 - Adding images to a web page
 - Image attributes: src ,alt, width and height
 - Using Images as Links
 - HTML Tables
 - Using Tables
 - Spanning Cells and Table Caption

Images

- Images can improve the design and appearance of a web page.
- Images appear on web page in two ways:
 - embedded in the inline content
 - as background images
- Main image formats that are used on the web:
 - GIF, JPEG and PNG

<https://www.pagecloud.com/blog/web-images-png-vs-jpg-vs-gif-vs-svg>

 element

- Syntax

``

- `src` attribute

- specifies the `URL` of the image file.

- `alt` attribute

- provides `alternate text` that displays if the image is not available.

 element

- It is an **empty element**.
- use **** in the stricter XHTML syntax.
- It is an **inline element**. Thus, it behaves like any other inline element in text flow.
- It is known as a **replaced element**, which is replaced by an external file when the page is displayed.
- By default, the bottom edge of an image aligns with the baseline of text.
- **Using CSS**, the author can float the image to the right or left margin, control the space and borders around the image, and change its vertical alignment.

 element

<p>Lavandula (common name lavender) is a genus of 47 known species of flowering plants in the mint family, Lamiaceae.</p>



Lavandula (common name lavender) is a genus of 47 known species of flowering plants in the mint family, Lamiaceae.

Demo

src attribute

- `src="URL"`
- ``

The image is in the same directory as the HTML document.

- ``

The image is in the directory /images.

- ``

The image is placed in other website.

alt attribute

- `alt="text"`
- The `alt` attribute must appear on every `<img>` element and its value should be a text description of the image.
- The important of the value of this attribute:
 - The alternative text will be shown if the browser cannot display the image.
 - For users with visual impairments, a screen reader software can read the alt text to describe the image that they cannot see.
 - The alternative text helps search engines index or find the contents of an image.
- If the image does not add anything meaningful to the text content of the page, the `alt` attribute **should still be used** and but **given no value**.
- If there is no alt attribute on the `<img>` element, the screen reader will announce the name of the image file, which may not make any sense to the user.

``

alternate text

Browsers handle the missing image

when you , the world smiles with you.

Firefox

when you smile, the world smiles with you.

Safari

when you , the world smiles with you.

Chrome

when you  smile, the world smiles with you.

Demo

width, height attributes

- specify the **width** and **height** of the image in pixels.
- can be a **percentage** of the browser screen such as "50%"
- Specifying the size of image is considered as good practice:
 - It helps a pages to load faster and smoothly.
 - Browser knows how much space to allocate to the image and can render correctly the rest of the page while the image is still loading.

Tips of image size

- Do not display images smaller or larger than they are.
 - Create version of images at the same size as them on the web pages.
- Do not use images that are bigger than they are shown on screen.
 - The smaller an image, the smaller the size of the file, the quicker the images loads in the browser.
- Do not specify the width or height attribute and leave out the other one
 - The browser will show the image to scale.

Different Image Dimensions



Fixed size width 50 height 50



Enlarged size width 100 height 100



Stretched size width 100 height 50



Stretched size width 50 height 100

Demo

loading attribute

- The loading attribute specifies whether a browser should load an image or to defer loading of off-screen images until for example the user scrolls near them.
- Tips: add loading="lazy" only images which are positioned below the fold.
- Syntax

``

- eager -- default. Loads an image immediately
- lazy -- defer loading of images until some conditions are met

Using Images as Links

- To use an image as a link, put the `<img>` element inside the `<a>` element

```
<a href="https://www.sit.kmutt.ac.th">  
    
</a>
```



Demo

Tables

- HTML tables represents **tabular data**
 - Data is presented in a two-dimensional table comprised of **rows** and **columns** of cells containing data.
- A **table cell** may contain any sort of information:
 - numbers
 - text elements
 - images
 - multimedia objects (audio, video)

Table Samples

Id ▴ ▾	Name ▴ ▾	Email ▴ ▾	Investments ▴ ▾	Action
261	Alfred Alan	aalan@gmail.com	Stocks	 
227	Alison Smart	asmart@biztalk.com	Residential Property	 
246	Ally Emery	allye@easymail.com	Stocks	 
212	Andrew Phips	andyp@mycorp.com	Stocks	 
218	Andy Mitchel	andym@hotmail.com	Stocks	 
221	Ann Melan	ann_melan@iinet.com	Residential Property	 
243	Ben Bessel	benb@hotmail.com	Stocks	 
232	Bensen Romanolf	benr@albert.net	Bonds	 
233	Brad Cole	bradc@hotmail.com	Stocks	 
241	Catherine Benchman	cathb@hotmail.com	Stocks	 

74 items found, displaying 1 to 10.

[First/Prev] [1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#) [Next/Last]

	BASIC	PROFESSIONAL	BUSINESS	ENTERPRISE	PREMIER
	\$24 monthly	\$59 monthly	\$99 monthly Most Popular	\$249 monthly	\$699 monthly
	Sign Up	Sign Up	Sign Up	Sign Up	Sign Up
Set up fee	waived	waived	waived	waived	waived
Transaction Fee	2.0%	1.0%	1.0%	0.5%	0%
Max SKUs	100	2,500	10,000	25,000	50,000
Storage	100 MB	500 MB	1000 MB	2.5 GB	5 GB
Bandwidth	unlimited	unlimited	unlimited	unlimited	unlimited
Custom Domains	✓	✓	✓	✓	✓
SSL Checkout	✓	✓	✓	✓	✓
SSL Admin		✓	✓	✓	✓
Discount Codes		✓	✓	✓	✓
Carrier Shipping			✓	✓	✓
Real-Time Stats			✓	✓	✓
Adwords credit	\$ 25.00	\$ 50.00	\$ 50.00	\$ 75.00	\$ 75.00
All prices are USD	Sign Up	Sign Up	Sign Up	Sign Up	Sign Up

Table Structure

Diagram illustrating the structure of a table with annotations:

- table**: Points to the entire table structure.
- table header**: Points to the first row (header row).
- row**: Points to individual data rows.
- table data**: Points to the data rows (excluding the header).
- column**: Points to individual columns.

Id ▾	Name ▾	Email ▾	Investments ▾	Action
261	Alfred Alan	aalan@gmail.com	Stocks	 
227	Alison Smart	asmart@biztalk.com	Residential Property	 
246	Ally Emery	allye@easymail.com	Stocks	 
212	Andrew Phips	andyp@mycorp.com	Stocks	 
218	Andy Mitchel	andym@hotmail.com	Stocks	 
221	Ann Melan	ann_melan@iinet.com	Residential Property	 
243	Ben Bessel	benb@hotmail.com	Stocks	 
232	Bensen Romanolf	benr@albert.net	Bonds	 
233	Brad Cole	bradc@hotmail.com	Stocks	 
241	Catherine Benchman	cathb@hotmail.com	Stocks	 

74 items found, displaying 1 to 10.
[First/Prev] 1, 2, 3, 4, 5, 6, 7, 8 [Next/Last]

Basic Table

Table

```
<table>
  <tr>
    <th>Menu item</th>
    <th>Calories</th>
    <th>Fat(g)</th>
  </tr>
  <tr>
    <td>Chichen noodle soup</td>
    <td>120</td>
    <td>2</td>
  </tr>
  <tr>
    <td>Caesar salad</td>
    <td>400</td>
    <td>26</td>
  </tr>
</table>
```

Menu item	Calories	Fat(g)
Chichen noodle soup	120	2
Caesar salad	400	26

Demo

Using Tables

- A **table** is defined using the `<table>` element
 - contains a number of **table cells** (`<td>`, for "table data") which are organized into table rows (`<tr>`).
- **Table cells** which act as **column headers** or **rows headers** should use the `<th>` element (table header).
- **Table cells** can be merged using the **colspan** and **rowspan** attributes.

Using Tables

- Tables can be broken into sections using the following elements:
 - `<thead>` Table header
 - `<tbody>` Table body
 - `<tfoot>` Table footer
- Browser can use these elements to enable scrolling of table body independently of the header and footer.
- Also, when printing a large table that spans multiple pages, these elements can enable the table header and footer to be printed at the top and bottom of each page.
- A caption can be added to a table using the `<caption>` element.

Sections in Table

Table

```
<table>
  <thead>
    <tr>
      <th>Month</th>
      <th>Savings</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>January</td>
      <td>$100</td>
    </tr>
    <tr>
      <td>February</td>
      <td>$80</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td>Sum</td>
      <td>$180</td>
    </tr>
  </tfoot>
</table>
```

```
<style>
  table {
    width: 50%;
    text-align: left ;
  }
  th, td {
    padding: 10px ;
    border-bottom: 1px solid #ddd;
  }
</style>
```

Month	Savings
January	\$100
February	\$80
Sum	\$180

Demo

Spanning Cells

- Cell spanning
 - is the **stretching** of a cell to cover several **rows** or **columns**.
 - allows to create **complex table structures**, but it has the **side effect** of making the markup a difficult to keep track of.
- To make a header or data cell span by adding the **colspan** or **rowspan** attributes.

Column spans

```
<table>
  <tr>
    <th>Name</th>
    <th colspan="2">Telephone</th>
  </tr>
  <tr>
    <td>Bill Gates</td>
    <td>55577854</td>
    <td>55577855</td>
  </tr>
</table>
```

Name	Telephone	
Bill Gates	55577854	55577855

Demo

Row spans

```
<table>
  <tr>
    <th>Name:</th>
    <td>Bill Gates</td>
  </tr>
  <tr>
    <th rowspan="2">Telephone:</th>
    <td>55577854</td>
  </tr>
  <tr>
    <td>55577855</td>
  </tr>
</table>
```

Name:	Bill Gates
Telephone:	55577854
	55577855

Demo

Table Caption

```
<table>
  <caption>Monthly savings</caption>
  <tr>
    <th>Month</th>
    <th>Savings</th>
  </tr>
  <tr>
    <td>January</td>
    <td>$100</td>
  </tr>
  <tr>
    <td>February</td>
    <td>$50</td>
  </tr>
</table>
```

Month	Savings
January	\$100
February	\$50

Demo

Common Table Mistakes

- Not closing the `<table>`, `<tr>` or `<td>` tags or closing them improperly.
- Inserting `<td>`'s outside of a `<tr>`
- Creating tables with differing numbers of cells (or `rowspan/colspan`) in each row
- Placing tables within inline elements (`<b>`) or `<h1>`
- Surrounding table cells or rows with text formatting tags

i.e. `<table><b><tr><td>Table Header</td></tr></b></table>`

- Data tables should have a caption, immediately after the opening table tag

`<table><caption>Table Caption</caption><tr> ...`

References

- https://www.w3schools.com/html/html_images.asp
- <https://www.pagecloud.com/blog/web-images-png-vs-jpg-vs-gif-vs-svg>
- https://www.w3schools.com/html/html_tables.asp
- <https://html.com/tables/>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/table>
- <https://www.freecodecamp.org/news/best-image-format-for-web-in-2019-jpeg-webp-heic-avif-41ba0c1b2789/>